

Chapter 6

Patient Safety and Quality Improvement (PSQ)

Intent of the chapter

The standards encourage an environment of patient safety and continual quality improvement. The patient safety and quality programme should be documented and involve all areas of the organisation and all staff members.

The management creates a culture of safety in the organisation. Patient safety officer(s) shall be designated for the implementation of patient-safety programme.

National / International patient-safety goals / solutions / framework shall be implemented.

The organisation shall collect data on structures, processes and outcomes, especially in areas of high-risk situations. Quality indicators specific to clinical specialities shall be used to involve clinical departments in the quality improvement programme. The collected data shall be collated, analysed and trends are used for further improvement. Appropriate quality tools shall be used for carrying out quality improvement projects. Clinical audits shall be used as a tool to improve the quality of patient care in a sustained manner. Department leaders play an active role in patient safety and quality improvement.

The organisation has a mechanism to capture patient reported outcome measures.

The organisation shall have a robust incident reporting system. Sentinel events shall be defined. All incidents are investigated and appropriate action is taken.

The management shall support the patient safety and quality programme.

SUMMARY OF STANDARDS

PSQ.1.	The organisation implements a structured patient safety programme.
PSQ.2.	The organisation implements a structured quality improvement and continuous monitoring programme.
PSQ.3.	The organisation identifies key indicators to monitor the structures, processes and outcomes, which are used as tools for continual improvement.
PSQ.4.	The organisation uses appropriate quality improvement tools for its quality improvement activities.
PSQ.5.	There is an established system for clinical audit.
PSQ.6.	The patient safety and quality improvement programme are supported by the management.
PSQ.7.	Incidents are collected and analysed to ensure continual quality improvement.

*This implies that the objective element requires documentation.

Summary of Objective Elements

Standard	7
Objective elements	46
CORE	8
Commitment	28
Achievement	7
Excellence	3

Objective Element	PSQ.1.	PSQ.2.	PSQ.3.	PSQ.4.	PSQ.5.	PSQ.6.	PSQ.7.
a	CORE	CORE	Commitment	CORE	Commitment	Achievement	CORE
b	Commitment	Commitment	CORE	Achievement	Commitment	Commitment	Commitment
c	Commitment	Excellence	Commitment	Commitment	Achievement	Commitment	Commitment
d	Commitment	Excellence	CORE	Achievement	Commitment	Commitment	Commitment
e	Commitment	Commitment	Commitment		Commitment	Achievement	Achievement
f	Commitment	Commitment	Commitment		Commitment	Excellence	Commitment
g	CORE	Commitment	Commitment				
h		Commitment	Achievement				
i		CORE					

Standards and Objective Elements

Standard

PSQ.1.

The organisation implements a structured patient safety programme.

Objective Elements

CORE

- a. The patient safety programme is developed, implemented and maintained by a multi-disciplinary safety committee. *

Interpretation: The safety committee must include representatives from administration, facility management, safety officers (patient and others), clinicians, nurses, paramedical staff and various clinical and support departments of the organisation. The organisation ensures that the safety committee meets every month to co-ordinate development, implementation and monitoring of the safety plans and policies to provide a safe and secure facility and environment. The committee shall ensure that patients are protected from harm either from the environment or from lack of appropriate care or lack of safety measures. The plans are implemented and reviewed in a monthly meeting.

The committee is responsible for proactive risk assessment. The same shall be done using tools like Hazard Identification and Risk Analysis (HIRA), Failure Modes and Effects Analysis (FMEA) in both clinical and non-clinical processes and areas.

The patient safety programme is documented as a manual. The manual shall incorporate all the requirements of this standard. This shall be documented, keeping in mind the requirements of other objective elements in the standard.

Refer to the glossary for a definition of "safety programme".

Refer to HRM.6.a. and PRE.5.j

- Commitment** b. The patient safety programme is comprehensive and covers all the major elements related to patient safety.

Interpretation: The patient safety programme shall address all elements of safety for clinical and support services.

Refer to AAC.7.e. AAC.7.f, AAC.7.g., AAC.9.f., AAC.9.g., AAC.9.h. AAC.9.i., AAC.9.j., AAC.9.k. and ROM.6.

Commitment c. The programme covers incidents ranging from “no harm” to “sentinel events”.

Interpretation: The organisation shall clearly define as to what constitutes “no harm” and “sentinel events” with regards to the patient.

Refer to the glossary for a definition of “no harm” and “sentinel events”.

Commitment d. Designated patient safety officer(s) coordinates implementation of the patient safety programme.

Interpretation: The patient safety officer(s) shall be a person having a good knowledge of both patient and general safety. The patient safety officer(s) shall report to the Top Management. The role and responsibilities of the patient safety officer(s) shall be defined and documented. The patient safety officer shall be a doctor/nurse with experience in the application of risk management to clinical domains. Also, champions in patient safety are identified and developed across the organisation and are supported to drive patient safety.

Commitment e. The organisation performs proactive analysis of patient safety risks and makes improvements accordingly.

Interpretation: Proactive analysis could be done using tools like Hazard Identification and Risk Analysis (HIRA), Failure Modes and Effects Analysis (FMEA), Hazard Vulnerability Analysis (HVA), fault tree analysis and simulations. At a minimum, one patient safety related risk shall undergo proactive analysis every year, and appropriate improvements are made as a result. The intent is to eliminate unsafe actions and conditions that can lead to harmful incidents.

Commitment f. The patient safety programme is reviewed and updated at least once a year.

Interpretation: The update shall be done based on newer literature on patient safety and the findings of reviews.

In case the annual review does not identify any opportunities for improvement, the same shall be documented in the minutes of the safety committee meeting.

The patient safety programme identifies opportunities for improvement based on the review at pre-defined intervals (as defined by the organisation in the Safety manual but at least once in a year). The review shall be done by safety committee and at a minimum includes report of facility inspection rounds, patient safety incidents, risk management and analysis of key-safety indicators. The minutes of the review meetings shall be recorded and maintained.

CORE

- g. The organisation adapts and implements national / international patient-safety goals / solutions / framework.**

Interpretation: At a minimum, the organisation shall adhere to the current National Patient-Safety Framework, WHO Patient Safety Solutions and/or International Patient Safety Goals.

Refer to AAC.10.d., COP.1.b., COP.7.d., COP.14.d., MOM.3.c., MOM.3.e., MOM.4.e., MOM.6.e., MOM.7.h., IPC.3.b. and IPC.3.d.

Standard

PSQ.2.

The organisation implements a structured quality improvement and continuous monitoring programme.

Objective Elements

CORE

- a. The quality improvement programme is developed, implemented and maintained by a multi-disciplinary committee. ***

Interpretation: The quality improvement programme shall be developed, implemented and maintained in a structured manner.

It shall be integrated across the organisation and provide a framework for risk management, ongoing monitoring and performing improvements based on reviews.

The roles and responsibilities of the multidisciplinary committee are defined, and it shall have representation from management, various clinical and support departments of the organisation. This committee shall receive inputs on significant deliberations from other committees in the organisation. The committee may be called as the core committee, quality improvement committee etc.

Commitment

- b. The quality improvement programme is comprehensive and covers all the major elements related to quality assurance. ***

Interpretation: The quality improvement programme shall be documented in the form of Quality Improvement Manual. The quality improvement programme shall incorporate the goals and objectives of the programme, framework for performing quality improvement activities including data collection, important

indicators as identified, frequency of mock drills, audit schedules, committees and their terms of reference, review policy and implementation of corrective and preventive action. Utilisation review preferably could be part of quality improvement program. The quality assurance programme for specific areas like Laboratory, Imaging, Emergency, Blood center, Operation theatre and the Intensive Care Unit(s) is also summarised.

Refer to AAC.7.a., AAC.9.a., COP.2.h., COP.8.g., COP.9.f., COP.14.i. COP.14.j. and PSQ.2.d.

Excellence **c. The quality improvement programme improves process efficiency and effectiveness.**

Interpretation: The quality improvement programme encourages the use of quality tools, novel strategies to improve both clinical and managerial processes. The impact of the managerial process innovations may be at the level of the department or organisation-wide. Innovations may be targeted to improve patient safety, improve care delivery, to reduce costs, to introduce environmentally-friendly measures etc. The management of the organisation promotes these innovations.

Excellence **d. The quality improvement programme focuses on appropriateness of clinical care.**

Interpretation: The doctors and nurses participate in utilization review which includes appropriateness of length of the stay, procedures, investigations and treatment.

Commitment **e. There is a designated individual for coordinating and implementing the quality improvement programme.***

Interpretation: The designated individual shall undergo a structured training programme in the implementation of accreditation standards (NABH Programme on Implementation of NABH Standards) and with knowledge of statutory requirements, hospital quality improvement principles and evaluation methodologies, hospital functioning and operations. For example, accreditation coordinator, quality management representative, quality manager.

The designated individual shall report to the Top Management. The role and responsibilities of the designated individual shall be defined and documented.

Also, champions in quality improvement are identified and developed across the organisation and supported to drive improvement.

Commitment f. The quality improvement programme identifies opportunities for improvement based on the review at pre-defined intervals.*

Interpretation: The quality improvement programme is a dynamic process. There is an outline of the periodic review mechanisms at different levels, such as department/senior administration/management reviews, etc. The Quality improvement programme needs to be reviewed by the quality improvement committee at regular pre-defined intervals as defined by the organisation in the quality improvement manual but at least once in three months. The review shall include findings of audits, organisational performance, analysis of key indicators as identified and determined by the organisation. The minutes of the review meetings shall be recorded and maintained.

Commitment g. The quality improvement programme is reviewed and updated at least once a year.

Interpretation: The update shall be done based on newer literature on quality improvement based on audits, feedback mechanisms, the review carried out by the quality improvement committee, etc.

In case the annual review does not identify any opportunities for improvement, the same shall be documented in the minutes of the quality improvement committee meeting.

Commitment h. Audits are conducted at regular intervals as a means of continuous monitoring.*

Interpretation: Choice and frequency of audits shall be defined for priority areas in the organisation and for areas of concern as identified by trends in key performance indicators, identified risk, etc. However, all the areas of the organisation shall be covered by a hospital-wide internal audit at least once in 6 months as per a scheduled plan. This internal audit shall be done by an identified staff or a multidisciplinary team trained in NABH Accreditation standards for Hospitals. They shall assess areas independent of their area of work. The internal audit of a particular area shall include all the applicable standards and objective elements of NABH Accreditation standards for Hospitals. At the end of the audit, there shall be a formal meeting to summarise the findings and corrective and preventive measures shall be taken and documented. Implementation of changes is verified and recorded.

CORE

- i. There is an established process in the organisation to monitor and improve the quality of nursing care.***

Interpretation: Monitoring shall be done through nursing audits. This could also be in the form of a competency evaluation by written questionnaire or witnessed demonstration for key nursing procedures such as medication administration, peripheral and intravenous cannula placement, tracheostomy care etc. The organisation shall identify key performance indicators that reflect excellence in nursing care. This shall be integrated with quality improvement programme of the organisation.

Standard**PSQ.3.**

The organisation identifies key indicators to monitor the structures, processes and outcomes, which are used as tools for continual improvement.

Objective Elements

- Commitment a. The organisation identifies and monitors key indicators to oversee the clinical structures, processes and outcomes.**

Interpretation: The organisation identifies and monitors the priority aspects of its patient care. Any indicator mandated by the Government of India/ State Government or the National Accreditation Board for Hospitals and Healthcare Providers (NABH) shall be monitored.

Some of the indicators that could be monitored pertain to patient assessment, safety and quality-control programmes of all the diagnostic services, medication management, use of blood and blood components, surgical services including use of anaesthesia and mortality and morbidity indicators.

Quality indicators pertaining to clinical specialties and sub specialties have been added in the Key Performance Indicators for greater involvement of clinicians in the quality improvement program. Few of these indicators have been mandatory for the organisation as per the scope of services.

Refer to NABH Key Performance Indicators.

These indicators are based on available literature or created in accordance with good practice. Every indicator shall have a defined numerator, denominator and multiplier. Where appropriate, the definition of terms shall be provided.

CORE

b. The organisation identifies and monitors the key indicators to oversee infection prevention and control activities.

Interpretation: The organisation shall identify and monitor appropriate key performance indicators suitable to it. Any indicator mandated by the Government of India or the National Accreditation Board for Hospitals and Healthcare Providers (NABH) shall be monitored.

Some of the indicators that could be monitored include the rate of Catheter Associated Urinary Tract Infection (CAUTI), Central Line Associated Blood Stream Infection (CLABSI), Surgical Site Infection (SSI) and Ventilator Associated Pneumonia (VAP).

These indicators are based on available literature or created in accordance with good practice. Every indicator shall have a defined numerator, denominator and multiplier. Where appropriate, the definition of terms shall be provided. In the case of healthcare associated infections, the current surveillance definitions provided by the CDC's National Healthcare Safety Network definitions shall be used.

Commitment c. The organisation identifies and monitors key indicators to oversee the managerial structures, processes and outcomes.

Interpretation: The organisation identifies and monitors priority managerial activities in the organisation. Any indicator mandated by the Government of India/ State Government or the National Accreditation Board for Hospitals and Healthcare Providers (NABH) shall be monitored.

Some of the indicators that could be monitored pertaining to medication procurement, utilisation rates, patient and staff satisfaction, waiting time for consultation and diagnostics, and availability and content of medical records.

Two indicators pertaining to digital health have been added.

These indicators are based on available literature or created in accordance with good practice. Every indicator shall have a defined numerator, denominator and multiplier. Where appropriate, the definition of terms shall be provided.

CORE

d. The organisation identifies and monitors key indicators to oversee patient safety activities.

Interpretation: The organisation shall identify and monitor appropriate key performance indicators suitable to it. Any indicator mandated by the Government of India / State Government or the National Accreditation Board for Hospitals and Healthcare Providers (NABH) shall be monitored.

Some of the indicators that could be monitored pertain to patient safety goals and risk management.

These indicators are based on available literature or created in accordance with good practice. Every indicator shall have a defined numerator, denominator and multiplier. Where appropriate, the definition of terms shall be provided.

Commitment e. Verification of data is done regularly by the quality team.

Interpretation: The data which is collected is verified from time to time and in response to queries or when an unexplained trend occurs, etc.

The data collected shall be authenticated by the quality team after going through every data or by using random sampling. The verification process shall also include if the stated sampling strategy (where applicable) was adhered.

Whenever errors are detected in the process of collection of data, they are corrected.

Commitment f. There is a mechanism for analysis of data which results in identifying opportunities for improvement.

Interpretation: The data is analysed and based on this corrective and preventive actions are taken, where necessary. The organisation could also consider developing benchmarks/acceptable quality levels based on national/international norms. This also includes undertaking quality improvement projects when the benchmark/acceptable quality levels are consistently not met, or the trend is unfavourable.

In case the organisation is consistently meeting benchmark/acceptable quality levels, they could reduce the frequency of monitoring.

Commitment g. The improvements are implemented and evaluated.

Interpretation: The improvement activities carried out by the organisation shall have an evaluable outcome. The same shall be documented.

Achievement h. Feedback about care and service is communicated to staff.

Interpretation: The feedback shall include the rates, trends and opportunities for improvement. It could also provide specific inputs to improve/reduce the rate. This could be in the form of a bulletin/newsletter. It is equally important that positive feedback about care and service is communicated to staff.

Standard

PSQ.4.

The organisation uses appropriate quality improvement tools for its quality improvement activities.

Objective Elements

CORE

a. The organisation undertakes quality improvement projects.

Interpretation: At a minimum, every year, the organisation shall undertake two quality improvement projects.

A project shall have a definite purpose, beginning and end. It shall be a time-bound activity. There shall be a measurement of parameters under improvement at the beginning and end of the project.

Achievement b. The Quality improvement projects shall include improvements in patient care delivery and hospital operations which will have an impact on cost and efficiency.

Interpretation: Quality improvement projects shall be centered on six domains of health care quality which include patient safety, cost effectiveness, patient centered, timeliness, efficiency and equity in healthcare.

Commitment c. The organisation uses appropriate analytical managerial and statistical tools for its quality improvement activities.

Interpretation: Depending on the project, the organisation shall identify and use appropriate analytical tools. For example, root cause analysis, run chart,

stratification diagram, normalisation table, flow chart, Pareto analysis, histogram etc.

Depending on the project, the organisation shall identify and use appropriate statistical tools such as sampling, control chart, ANOVA, six sigma etc.

Depending on the project, the organisation shall identify and use appropriate managerial tools. For example, Lean, Six Sigma, Plan Do Study Act (PDSA), Force Field Analysis, Affinity Diagram, Matrix Diagram, Tree Diagram, etc.

Achievement d. The organisation has a mechanism to capture patient reported outcome measures.

Interpretation: The organisation shall develop a mechanism to capture patient reported outcome measures (PROMs). The organisation needs to ensure that the measurement instruments that patients complete to provide information on aspects of their health status incorporate information that is relevant to their quality of life, including symptoms, functionality and physical, mental and social health. PROMs can be classified as either generic or disease-specific. The organisation shall use PROMs to better communication and decision making between doctors and patients and improve patient satisfaction with care.

Standard

PSQ.5. There is an established system for clinical audit.

Objective Elements

Commitment a. Clinical audits are performed to improve the quality of patient care.

Interpretation: The organisation shall use clinical audits as a quality improvement tool to improve the quality of patient care. The clinical audit could be retrospective/prospective in nature. The topic for audit could be disease-based, cost-based, community-based or based on morbidity (length of stay). The organisation shall conduct one clinical audit per clinical department once in two years as per the scope of services. The organisation needs to take care to differentiate clinical audit from research projects.

Commitment b. The parameters to be audited are defined by the organisation.

Interpretation: As clinical audits are standards-based, they must be done using predefined parameters so that there is no bias. The organisation shall lay down the objectives, the standards against which the audit will be conducted, develop a checklist where required, sampling and data collection guidelines and preparation of the report. The audit shall encompass aspects of clinical and nursing care.

Achievement c. Medical and nursing staff participate in clinical audit.

Interpretation: The organisation shall identify such personnel. It could be a mix of clinicians, administrators and nurses. These could be members of the core committee/quality assurance committee etc.

Commitment d. Patient and staff anonymity are maintained.

Interpretation: This means that the names of the patients and the hospital staff who may figure in the audit documents must not be disclosed nor any reference is made to them in public discussions/conferences. This is at the stage of report preparation and dissemination. The staff participating in the audit shall maintain patient and staff anonymity and not reveal names.

Commitment e. Clinical audits are documented.

Interpretation: The organisation could use a checklist with the predefined parameters, and the audit findings could be recorded on this sheet. After the audit, a report shall be prepared, highlighting the key findings of the audit.

Commitment f. Remedial measures are implemented.

Interpretation: All remedial measures as ascertained shall be documented and implemented, and improvements thereof recorded to complete the audit cycle. This shall preferably be done based on the root-cause analysis.

Standard

PSQ.6.

The patient safety and quality improvement programme are supported by the management.

Objective Elements

Achievement a. The management creates a culture of safety.

Interpretation: The management needs to ensure the adoption of behaviours that promote patient safety. Some of the key features required for a culture of safety are sharing information, reporting occurrences of incidents, learning from safety incident analysis, blame-free culture and encouragement of collaboration across disciplines and departments. The key components of patient safety culture are informed culture, reporting culture, learning culture, just culture and flexible culture.

The management needs to measure its safety culture regularly (at least once a year). This shall be measured using validated surveys. For example, the Manchester Patient Safety Framework (MaPSaF), Safety Attitudes Questionnaire, AHRQ Surveys on Patient Safety Culture (SOPS™). The management shall act on their patient safety culture assessment results.

Commitment b. The leaders at all levels in the organisation are aware of the intent of the patient safety and quality improvement programme and the approach to its implementation.

Interpretation: The organisation and departmental leaders are aware of the patient safety and quality improvement programme, its intent and applicability to the respective areas and how it contributes to the organisation as a whole.

Commitment c. Departmental leaders are involved in patient safety and quality improvement.

Interpretation: To effectively implement this, each department could have its department objectives/key performance indicators including clinical indicators, and the responsibility of achieving them could be that of the leader/designated person. This could be discussed during departmental meeting.

Commitment d. Organisation earmarks adequate funds from its annual budget in this regard.

Interpretation: Appropriate fund allocation is done by the organisation for the smooth functioning of the patient safety and quality improvement programme. The budget could be earmarked based on previous year's spending. If no data is available, the organisation could make a beginning by earmarking a budget but reviewing it at the end of six months to make any necessary modifications. The management makes available adequate resources required for patient safety and quality improvement programme. Resources shall include men, material, machine, money, milieu, measurement and method. These shall be in steady supply to ensure that the programme functions smoothly.

Achievement e. The management identifies organisational performance improvement targets.

Interpretation: The management shall identify the organisation and department level quality objectives, set targets, monitor them (at least once in three months) and modify the target (at least annually). The targets shall be shared with the faculty and staff and regular feedback taken.

Excellence f. The management uses the feedback obtained from the work force to improve patient safety and quality improvement programme.

Interpretation: The feedback shall be obtained from the staff on their understanding and use of the safety and quality systems. Feedback may be obtained once a year through staff surveys, but it is also important that staff workforce feel able to raise concerns whenever they occur. These inputs shall be used to improve patient safety and quality improvement programmes.

Standard

PSQ.7.

Incidents are collected and analysed to ensure continual quality improvement.

Objective Elements

CORE

a. The organisation implements an incident management system.*

Interpretation: The incident management system includes:

- identification
- reporting
- review
- action on incidents

This system supports factual reporting and learning and is based on the principle of just culture.

The organisation shall have a mechanism for reporting the occurrence of incidents on standardised incident report forms. It is preferable that the reporting system is simple (a few steps), clear (what needs to be reported, how to report, and to whom), confidential, and focused on process improvement.

While capturing the incidents, the organisation shall capture all incidents without going into the severity or whether harm was caused.

Commitment b. The organisation has a mechanism to identify sentinel events.*

Interpretation: The sentinel events relating to system or process deficiencies that are relevant and important to the organisation must be clearly defined. The list of the identified and relevant sentinel events shall be documented.

Refer to the glossary for a definition of "sentinel events".

Commitment c. The organisation has established processes for analysis of incidents.

Interpretation: The safety committee (Refer to PSQ.1.a.) shall be responsible for this activity. This could preferably be done by identifying the root cause. Inputs could be sought from the units/discipline/departments concerned. Where possible, patients and other stakeholders could be included in analysing the feedback and complaint.

The immediate response to a safety incident shall be to address the urgent care and support needs of those involved. This shall not await analysis

In case of sentinel events, correction if any shall be initiated within 24-working hours of occurrence or reporting. The analysis of sentinel events shall be completed within seven working days of occurrence or reporting.



CORE



Commitment



Achievement



Excellence

Commitment d. Corrective and preventive actions are taken based on the findings of such analysis.

Interpretation: The objective of this is to improve the quality of patient-care services continually. All such action shall be documented. The findings and recommendations arrived at after the analysis shall be communicated to all personnel concerned to correct the systems and processes to prevent recurrences. Any change in the policy or procedure is reflected as an amendment in the organisation's documentation.

Achievement e. The organisation incorporates risks identified in the analysis of incidents into the risk management system.

Interpretation: If the incident analysis reveals the presence of an unidentified risk, the same shall be subjected to risk management.

Commitment f. The organisation shall have a process for informing various stakeholders in case of a near miss / adverse event / sentinel event.

Interpretation: After due analysis, an incident could be termed as a near miss or adverse event or sentinel event. Based on the nature of the near miss or adverse event or sentinel event, the organisation shall inform the stakeholders (including patient and family where applicable) regarding the relevant concerns in addition to initiating the corrective and preventive action.

References

1. Leotsakos A, Zheng H, Croteau R, Loeb JM, Sherman H, Hoffman C, Morganstein L, O'leary D, Bruneau C, Lee P, Duguid M. Standardization in patient safety: the WHO High 5s project. *International journal for quality in health care*. 2014 Apr 1;26(2):109-16.
2. Chou D. Health IT and patient safety: building safer systems for better care. *Jama*. 2012 Dec 5;308(21):2282.
3. Frankel A, Federico F, Frush K, Haraden C, Kaplan GS. The essential guide for patient safety officers. Leonard M, editor. Oakbrook Terrace, IL: Joint Commission Resources [and] Institute for Healthcare Improvement; 2013.
4. World Health Organization. (n.d.). Patient Safety Solutions. [cited 2023 Dec 5]. Available from: <https://www.who.int/teams/integrated-health-services/patient-safety/research/patient-safety-solutions>
5. de Souza Gomes AP, Querido DL, da Silva GR, de Almeida LF, Rocha RG. Identificação do paciente em neonatologia para assistência segura. *Cogitare Enfermagem*. 2017 Aug 29;22(3).
6. ECRI Institute. ECRI Institute PSO Deep Dive: Patient Identification: Executive Summary. Aug 2016. [cited 2023 Dec 5]. Available from: https://www.ecri.org/Resources/Whitepapers_and_reports/PSO%20Deep%20Dives/Deep%20Dive_P_T_ID_2016_exec%20summary.pdf
7. Montgomery C, Wei E. Patient Identification. In *Patient Safety: A Case-based Innovative Playbook for Safer Care* 2023 Oct 17 (pp. 75-87). Cham: Springer International Publishing.
8. ECRI Institute. Patient Identification Errors. Jun 2016. [cited 2023 Dec 5]. Available from: https://www.ecri.org/Resources/HIT/Patient%20ID/Patient_Identification_Evidence_Based_Literature_final.pdf
9. ECRI Partnership for Health IT and Patient Safety. Health IT Safe Practices: Toolkit for the Safe Use of Health IT for Patient Identification. Feb 2017. [cited 2023 Dec 5]. Available from: https://www.ecri.org/Resources/HIT/Patient%20ID/Patient_Identification_Toolkit_final.pdf
10. Hughes P, Latino RJ, Kelly T. Preventing Patient Identification Errors.
11. Desai KN, Chaudhari S. Analysis of critical values in NABL (National Accreditation Board for Testing and Calibration Laboratories) accredited hematology and clinical pathology laboratory. *Ann Appl Bio-Sciences*. 2017 Feb 14;4(1):A14-8.
12. Patton LJ, Tidwell JD, Falder-Saeed KL, Young VB, Lewis BD, Binder JF. Ensuring safe transfer of pediatric patients: a quality improvement project to standardize handoff communication. *Journal of Pediatric Nursing*. 2017 May 1;34:44-52.
13. Starmer AJ, Schnock KO, Lyons A, Hehn RS, Graham DA, Keohane C, Landrigan CP. Effects of the I-PASS Nursing Handoff Bundle on communication quality and workflow. *BMJ quality & safety*. 2017 Dec 1;26(12):949-57.
14. Institute for Safe Medication Practices. ISMP List of High-Alert Medications in Acute Care Settings. 2018. [cited 2023 Dec 5]. Available from: <https://www.ismp.org/sites/default/files/attachments/2018-08/highAlert2018-Acute-Final.pdf>
15. World Health Organization. Medication safety in high-risk situations. World Health Organization; 2019.
16. Cabral RA, Eggenberger T, Keller K, Gallison BS, Newman D. Use of a surgical safety checklist to improve team communication. *AORN journal*. 2016 Sep 1;104(3):206-16

17. Jain D, Sharma R, Reddy S. WHO safe surgery checklist: Barriers to universal acceptance. *Journal of anaesthesiology, clinical pharmacology*. 2018 Jan;34(1):7.
18. World Health Organization. Surgical Safety Checklist. (Updated: Jan 1, 2009.) [cited 2023 Dec 5]. Available from: <https://www.who.int/teams/integrated-health-services/patient-safety/research/safe-surgery/tool-and-resources>.
19. Resar R, Griffin FA, Haraden C, Nolan TW. Using care bundles to improve health care quality. IHI innovation series white paper. Cambridge, Massachusetts: Institute for Healthcare Improvement. 2012;14:15-6.
20. World Health Organization. Infection Prevention and Control. Hand Hygiene: The Evidence for Clean Hands. [cited 2023 Dec 5]. Available from: <https://www.who.int/teams/integrated-health-services/infection-prevention-control/hand-hygiene>
21. Lotfinejad N, Peters A, Tartari E, Fankhauser-Rodriguez C, Pires D, Pittet D. Hand hygiene in health care: 20 years of ongoing advances and perspectives. *The Lancet infectious diseases*. 2021 Aug 1;21(8):e209-21.
22. Agency for Healthcare Research and Quality. Preventing Falls in Hospitals: A Toolkit for Improving Quality of Care. AHRQ Publication No. 12-0015-EF. Jan 2013. [cited 2023 Dec 5]. Available from: <https://www.ahrq.gov/patient-safety/settings/hospital/fall-prevention/toolkit/index.html>
23. France D, Slayton J, Moore S, Domenico H, Matthews J, Steaban RL, Choma N. A multicomponent fall prevention strategy reduces falls at an academic medical center. *The Joint Commission Journal on Quality and Patient Safety*. 2017 Sep 1;43(9):460-70.
24. World Health Organization. Falls. 26 April, 2021. [cited 2023 Dec 5]. Available from: <https://www.who.int/news-room/fact-sheets/detail/falls>
25. Donabedian A. Quality assessment and monitoring: retrospect and prospect. *Evaluation & the health professions*. 1983 Sep;6(3):363-75.
26. Duffy JR. Quality caring in nursing and health systems: Implications for clinicians, educators, and leaders. Springer Publishing Company; 2022 Dec 22.
27. Zarei E, Karimi S, Mahfoozpour S, Marzban S. Assessing hospital quality management systems: evidence from Iran. *International journal of health care quality assurance*. 2019 Feb 11;32(1):87-96.
28. Lee SE, Scott LD, Dahinten VS, Vincent C, Lopez KD, Park CG. Safety culture, patient safety, and quality of care outcomes: a literature review. *Western journal of nursing research*. 2019 Feb;41(2):279-304.
29. Jones P, Shepherd M, Wells S, Le Fevre J, Ameratunga S. What makes a good healthcare quality indicator? A systematic review and validation study. *Emergency Medicine Australasia*. 2014 Apr;26(2):113-24.
30. Kötter T, Blozik E, Scherer M. Methods for the guideline-based development of quality indicators--a systematic review. *Implementation Science*. 2012 Dec;7:1-22.
31. Dimick JB. What makes a "good" quality indicator?: comment on "Quality indicators for evaluating trauma care". *Archives of Surgery*. 2010 Mar 1;145(3):295-.
32. Doyle C, Lennox L, Bell D. A systematic review of evidence on the links between patient experience and clinical safety and effectiveness. *BMJ open*. 2013 Jan 1;3(1):e001570.
33. Fung CH, Lim YW, Mattke S, Damberg C, Shekelle PG. Systematic review: the evidence that publishing patient care performance data improves quality of care. *Annals of internal medicine*. 2008 Jan 15;148(2):111-23.

34. Gruen RL, Gabbe BJ, Stelfox HT, Cameron PA. Indicators of the quality of trauma care and the performance of trauma systems. *Journal of British Surgery*. 2012 Jan;99(Supplement_1):97-104.
35. US Centers for Disease Control and Prevention, National Healthcare Safety Network. NHSN Data Validation. (Updated: Apr 3, 2019.) [cited 2023 Dec 5]. Available from: <https://www.cdc.gov/nhsn/validation/index.html>
36. Jones P, Shepherd M, Wells S, Le Fevre J, Ameratunga S. What makes a good healthcare quality indicator? A systematic review and validation study. *Emergency Medicine Australasia*. 2014 Apr;26(2):113-24.
37. Kötter T, Blozik E, Scherer M. Methods for the guideline-based development of quality indicators--a systematic review. *Implementation Science*. 2012 Dec;7:1-22.
38. Catalyst NE. What is risk management in healthcare?. *NEJM Catalyst*. 2018 Apr 25;4(2).
39. Rubin HR, Pronovost P, Diette GB. The advantages and disadvantages of process-based measures of health care quality. *International Journal for Quality in Health Care*. 2001 Dec 1;13(6):469-74.
40. Santana MJ, Ahmed S, Lorenzetti D, Jolley RJ, Manalili K, Zelinsky S, Quan H, Lu M. Measuring patient-centred system performance: a scoping review of patient-centred care quality indicators. *BMJ open*. 2019 Jan 1;9(1):e023596.
41. Institute for Healthcare Improvement. (n.d.). Quality Improvement Essentials Toolkit [cited 2023 Dec 5]. Available from: <http://www.ihl.org/resources>
42. Hughes RG. Tools and strategies for quality improvement and patient safety. *Patient safety and quality: An evidence-based handbook for nurses*. 2008 Apr.
43. Smith DL. FMEA: preventing a failure before any harm is done. *isixsigma magazine*, march. 2004.
44. American Society for Quality. (n.d.). 7 Basic Quality Tools: Quality Management Tools. [cited 2023 Dec 5]. Available from: <https://asq.org/quality-resources/seven-basic-quality-tools>
45. Agency for Healthcare Research and Quality. (n.d.). Section 4: Ways To Approach the Quality Improvement Process (Page 1 of 2). [cited 2023 Dec 5]. Available from: <https://www.ahrq.gov/cahps/quality-improvement/improvement-guide/4-approach-qi-process/index.html>
46. Agency for Healthcare Research and Quality. (n.d.). TeamSteps: Strategies and Tools to Enhance Performance and Patient Safety. [cited 2023 Dec 5]. Available from: <http://www.ahrq.gov/professionals/education>
47. American Society for Quality. (n.d.). Quality Statistics - Statistical Methods for Quality Improvement. [cited 2023 Dec 5]. Available from: <https://asq.org/quality-resources/statistics>
48. American Society for Quality. (n.d.). What is Root Cause Analysis (RCA)? [cited 2023 Dec 5]. Available from: <https://asq.org/quality-resources/root%20cause%20analysis>
49. Quality Improvement Essentials Toolkit. (n.d.). Institute for Healthcare Improvement. [cited 2023 Dec 5]. Available from: <https://www.ihl.org/resources/tools>
50. Quality Statistics - Statistical Methods for Quality Improvement. (n.d.). American Society for Quality [cited 2023 Dec 5]. Available from: <https://asq.org/quality-resources/statistics>
51. Krause C. The case for quality improvement. *Healthc Q*. 2017 Jan 1;20(1):25-7.
52. Kingsley C, Patel S. Patient-reported outcome measures and patient-reported experience measures. *Bja Education*. 2017 Apr 1;17(4):137-44.

53. Gilmore KJ, Corazza I, Coletta L, Allin S. The uses of patient reported experience measures in health systems: a systematic narrative review. *Health policy*. 2023 Feb 1;128:1-0.
54. Monmouth Partners. A Guide to Patient Reported Measures – Theory, Landscape and Uses. [cited 2023 Dec 5]. Available from: https://silo.tips/download/a-guide-to-patient-reported-measures-theory-landscape-and-uses#google_vignette
55. What are PROMs? Centre for Health Services and Policy Research at University of British Columbia. [cited 2023 Dec 5]. Available from: <https://patientreportedoutcomes.ca/what-are-pros/proms/>
56. Bull C, Teede H, Watson D, Callander EJ. Selecting and implementing patient-reported outcome and experience measures to assess health system performance. In *JAMA Health Forum* 2022 Apr 1 (Vol. 3, No. 4, pp. e220326-e220326). American Medical Association.
57. Withers K, Palmer R, Lewis S, Carolan-Rees G. First steps in PROMs and PREMs collection in Wales as part of the prudent and value-based healthcare agenda. *Quality of life research*. 2021 Nov;30:3157-70.
58. Jonsson Å, Orwelius L, Dahlstrom U, Kristenson M. Evaluation of the usefulness of EQ-5D as a patient-reported outcome measure using the Paretian classification of health change among patients with chronic heart failure. *Journal of patient-reported outcomes*. 2020 Dec;4(1):1-1.
59. Bull C, Byrnes J, Hettiarachchi R, Downes M. A systematic review of the validity and reliability of patient-reported experience measures. *Health services research*. 2019 Oct;54(5):1023-35.
60. Kingsley C, Patel S. Patient-reported outcome measures and patient-reported experience measures. *Bja Education*. 2017 Apr 1;17(4):137-44.
61. Bull C, Latimer S, Crilly J, Gillespie BM. A systematic mixed studies review of patient experiences in the ED. *Emergency Medicine Journal*. 2021 Aug 1;38(8):643-9.
62. Churrua K, Pomare C, Ellis LA, Long JC, Henderson SB, Murphy LE, Leahy CJ, Braithwaite J. Patient-reported outcome measures (PROMs): a review of generic and condition-specific measures and a discussion of trends and issues. *Health Expectations*. 2021 Aug;24(4):1015-24.
63. Black N, Varaganum M, Hutchings A. Relationship between patient reported experience (PREMs) and patient reported outcomes (PROMs) in elective surgery. *BMJ quality & safety*. 2014 Jul 1;23(7):534-42.
64. Australian Commission on Safety and Quality in Healthcare. November 2016. Patient Reported Outcome Measures. Literature Review [cited 2023 Dec 5]. Available from: <https://www.safetyandquality.gov.au/sites/default/files/migrated/PROMs-Literature-Review-December-2016.pdf>
65. Patient reported outcome measures (PROMs) in England—2011-2012, special topic, EQ-5D™ Index analysis. NHS Digital. [cited 2023 Dec 5]. Available from: <https://digital.nhs.uk/data-and-information/publications/statistical/patient-reported-outcome-measures-proms/patient-reported-outcome-measures-proms-in-england-2011-2012-special-topic-eq-5d-index-analysis>
66. Weldring T, Smith SM. Article commentary: patient-reported outcomes (pros) and patient-reported outcome measures (PROMs). *Health services insights*. 2013 Jan;6:HSI-S11093.
67. Lagha E, Noble A, Smith A, Denvir MA, Leslie SJ. Patient Reported Experience Measures (PREMs) in chronic heart failure. *The Journal of the Royal College of Physicians of Edinburgh*. 2012 Jan 1;42(4):301-5.
68. Dawson J, Doll H, Fitzpatrick R, Jenkinson C, Carr AJ. The routine use of patient reported outcome measures in healthcare settings. *Bmj*. 2010 Jan 18;340.

69. Shariff R, AbuGhazaleh S, Sagr E, Ebdah M, Baumhauer J. Leveraging Patient-Reported Experience Measures (PREMs) Data to Deploy Patient-Reported Outcome Measures (PROMs). *NEJM Catalyst Innovations in Care Delivery*. 2022 Jun 15;3(7):CAT-22.
70. Hart AR. Principles for Best Practice in Clinical Audit: By NHS, National Institute for Clinical Excellence, CHI (Commission for Health Improvement), Royal College of Nursing, University of Leicester. (Pp 196, paperback plus CD ROM;£ 19.95 for NHS staff and organisations,£ 29.95 for those not working for the NHS.) Radcliffe Medical Press, 2002. ISBN 1-85775-976-1.
71. Limb C, Fowler A, Gundogan B, Koshy K, Agha R. How to conduct a clinical audit and quality improvement project. *IJS Oncology*. 2017 Jul 1;2(6):e24.
72. University Hospitals Bristol. (n.d.). How To Guides. [cited 2023 Dec 5]. Available from: <http://www.uhbristol.nhs.uk/for-clinicians/clinicalaudit/how-to-guides/>
73. Knight H, Gurol-Urganci I, Mahmood T. Clinical Audit for Quality Improvement. Introduction to Research Methodology for Specialists and Trainees. 2017 Aug 17:53. Knight H, Gurol-Urganci I, Mahmood T. Clinical Audit for Quality Improvement. Introduction to Research Methodology for Specialists and Trainees. 2017 Aug 17:53.
74. National Clinical Audit and Patient Outcomes Programme. Secondary National Clinical Audit and Patient Outcomes Programme 2014. [cited 2023 Dec 5]. Available from: www.hqip.org.uk/national-clinical-audits-managed-by-hqip
75. Healthcare Quality Improvement Partnership. / A-Z of National Clinical Audits. The National Clinical Audit Programme. [cited 2023 Dec 5]. Available from: <https://www.hqip.org.uk/a-z-of-nca/>.
76. Frankel A, Leonard M. 2013 Aug 22. The PS Net Collection. Perspectives. Update on Safety Culture. Agency for Healthcare Research and Quality. Patient Safety Network. [cited 2023 Dec 5]. Available from: <https://psnet.ahrq.gov/perspective/update-safety-culture>
77. Agency for Healthcare Research and Quality. Patient Safety Network. (2019, January). Culture of Safety. [cited 2023 Dec 5]. Available from: <http://psnet.ahrq.gov/primer.aspx?primerID=5>
78. Agency for Healthcare Research and Quality. SOPS Surveys. Hospital Survey on Patient Safety Culture [cited 2023 Dec 5]. Available from <https://www.ahrq.gov/sops/surveys/hospital/index.html>
79. The Health Foundation. (2012, May). How can leaders influence a safety culture? [cited 2023 Dec 5]. Available from : <https://www.health.org.uk/sites/default/files/HowCanLeadersInfluenceASafetyCulture.pdf>
80. Secanell M, Groene O, Arah OA, Lopez MA, Kutryba B, Pfaff H, Klazinga N, Wagner C, Kristensen S, Bartels PD, Garel P. Deepening our understanding of quality improvement in Europe (DUQuE): overview of a study of hospital quality management in seven countries. *International journal for quality in health care*. 2014 Apr 1;26(suppl_1):5-15.
81. Swensen SJ, Dilling JA, Mc Carty PM, Bolton JW, Harper Jr CM. The business case for health-care quality improvement. *Journal of patient safety*. 2013 Mar 1;9(1):44-52.
82. Fung CH, Lim YW, Mattke S, Damberg C, Shekelle PG. Systematic review: the evidence that publishing patient care performance data improves quality of care. *Annals of internal medicine*. 2008 Jan 15;148(2):111-23.
83. Thomas EJ. The future of measuring patient safety: prospective clinical surveillance. *BMJ Quality & Safety*. 2015 Apr 1;24(4):244-5.

84. Thomas L, Galla C. Republished: building a culture of safety through team training and engagement. *Postgraduate Medical Journal*. 2013 Jul;89(1053):394-401.
85. Trbovich PL, Griffin M. Measuring and improving patient safety culture: still a long way to go.
86. Tsai TC, Jha AK, Gawande AA, Huckman RS, Bloom N, Sadun R. Hospital board and management practices are strongly related to hospital performance on clinical quality metrics. *Health affairs*. 2015 Aug 1;34(8):1304-11.
87. Wagner C, Smits M, Sorra J, Huang CC. Assessing patient safety culture in hospitals across countries. *International Journal for Quality in Health Care*. 2013 Jul 1;25(3):213-21.
88. Weaver SJ, Lubomksi LH, Wilson RF, Pfoh ER, Martinez KA, Dy SM. Promoting a culture of safety as a patient safety strategy: a systematic review. *Annals of internal medicine*. 2013 Mar 5;158(5_Part_2):369-74.
89. Beard P, Greenall J, Hoffman CE, Nettleton S, Popescu IC, Ste-Marie M, Walsh D, White J. Canadian incident analysis framework.
90. Agency for Healthcare Research and Quality. Patient Safety Network. (2019, January). Root Cause Analysis. [cited 2023 Dec 5]. Available from: <https://psnet.ahrq.gov/primers/primer/10/Root-Cause-Analysis>
91. Agency for Healthcare Research and Quality. Patient Safety Network. (2019, January). Reporting Patient Safety Events. [cited 2023 Dec 5]. Available from: <https://psnet.ahrq.gov/primers/primer/13/reporting-patient-safety-events%20on%20April%2016>
92. Agency for Healthcare Research and Quality. Patient Safety Primers. (2019, January). Detection of Safety Hazards. [cited 2023 Dec 5]. Available from: <https://psnet.ahrq.gov/primers/primer/24/Detection-of-Safety-Hazards>
93. Beard P, Greenall J, Hoffman CE, Nettleton S, Popescu IC, Ste-Marie M, Walsh D, White J. Canadian incident analysis framework.
94. Charles R, Hood B, Derosier JM, Gosbee JW, Li Y, Caird MS, Biermann JS, Hake ME. How to perform a root cause analysis for workup and future prevention of medical errors: a review. *Patient safety in surgery*. 2016 Dec;10:1-5.
95. Ewen BM, Bucher G. Root cause analysis: Responding to a sentinel event. *Home Healthcare Now*. 2013 Sep 1;31(8):435-43.
96. Bagian JP, Bonacum D, DeRosier J, Frost J, Fairbanks RJ, Gandhi T, Sheppard F. RCA 2: Improving Root Cause Analyses and Actions to Prevent Harm
97. Carayon P. Human factors in patient safety as an innovation. *Applied ergonomics*. 2010 Sep 1;41(5):657-65.

Chapter 7

Responsibilities of Management (ROM)

Intent of the chapter

The management of the healthcare organisation is aware of and manages all the key components of governance. Those responsible for governance are identified and their roles defined. The standards encourage the governance of the organisation professionally and ethically.

Clinical governance framework is established, that includes clinical audits, clinical pathways, education and research. The responsibilities of management are defined. The responsibilities of the leaders at all levels are defined. The management executes its responsibility for compliance with all applicable regulations. Those responsible for governance address the organisations social responsibility.

Leaders ensure that patient-safety and risk-management issues are an integral part of patient care and hospital management. The Organisation has a written guidance in place for change management and service continuity plan.

Note 1: "Responsible for Governance" refers to the governing entity of the healthcare organisation and can exist in many configurations. For example, the owner(s), partners in partnership firm, trustees in charitable trust, the board of directors, or in the case of public hospitals, the respective Ministry (Health / Railways / Labour).

Note 2: "Leadership" refers to appointed leader for example CEO, COO, Managing Director, Dean / Director, Medical Director / Medical Superintendent.

In case of single owner / partners all the standards and objective elements shall be applicable.

SUMMARY OF STANDARDS

ROM.1.	The organisation identifies those responsible for governance and their roles are defined.
ROM.2.	Those responsible for governance manage the organisation in an ethical manner.
ROM.3.	Those responsible for governance ensure sustainability in hospital by addressing environmental, social and economic factors from long term well-being of healthcare system and community.
ROM.4.	The organisation is headed by a leader who shall be responsible for operating the organisation on a day-to-day basis.
ROM.5.	The organisation displays professionalism in its functioning.
ROM.6.	Management ensures that patient-safety aspects and risk-management issues are an integral part of patient care and hospital management.

*This implies that the objective element requires documentation.